

BECKHOFF

Special Projects Team

TwinCAT HMI Style Guide

v1.4

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Contents

Layout and Content.....	4
Header	4
Navigation.....	4
Machine Control Flyout.....	4
Default Pages	5
Home	6
Application Specific Pages.....	6
Manual.....	6
Recipes	6
Events	7
Diagnostics	8
Trends	8
EtherCAT	8
System	9
Settings Popup	9
Folder Layout	10
Responsive Design	10
Grouping of Controls	11
Styling.....	12
Control Default Styling.....	12
Colors	12
Font.....	14
Iconography	15
UI Components	16
Standard Button Setup and Databinding	16
Confirmation Dialog.....	17
Popups	17
Data Display Tiles	18

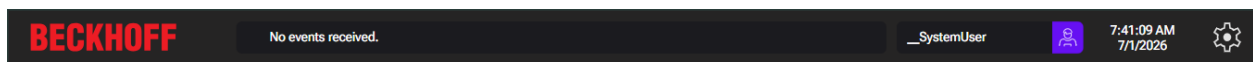
Setup and Conventions	18
Naming Conventions	18
Version Control	18
New Project Generation	19

Layout and Content

This section will provide details about the standard content in SPT HMIs and how it is laid out.

Header

A header is included at the top and displayed at all times. It contains a company logo, left justified, and an event line, settings button, date time display, and user management, right justified.



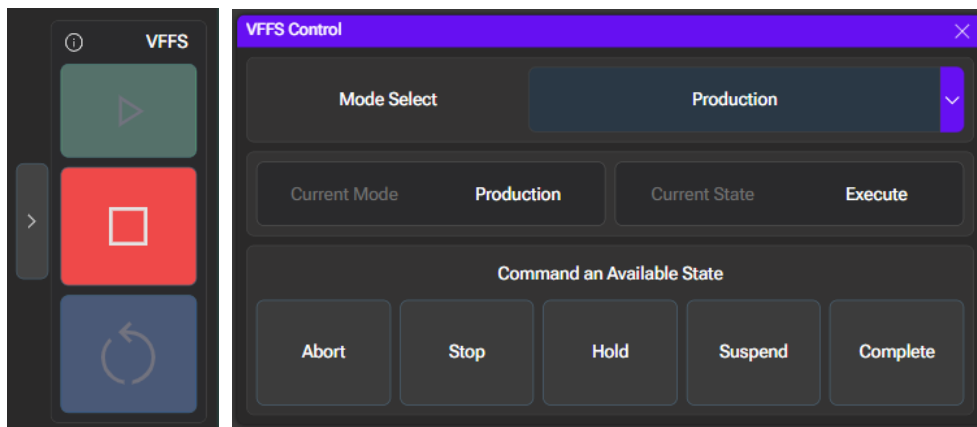
Navigation

Accordion Navigation is used and positioned on the left. Common iconography is used. Icons are SVGs with grayscale coloring. For consistency and quality, icons come from Google Fonts, as described in the [Iconography](#) section below.

The Back button is not used but can be included if navigation becomes complex. In general, navigation should be simple and intuitive.

Machine Control Flyout

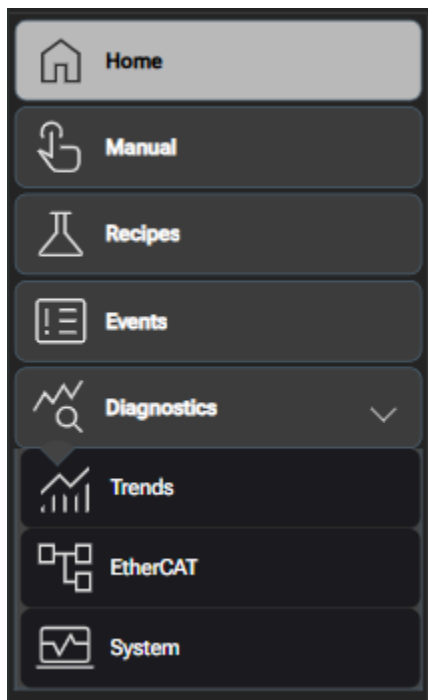
Contains control for machine start, stop, and reset, as well as the ability to open a popup for PackML Mode and State control. The flyout can be hidden, opened, and dragged to move up and down.



Default Pages

Default Pages include the following, with any application specific pages going between Home and Manual. Nested pages are indicated as indented bullets –

- Home
- Application Specific Pages (not included in template)
- Manual
- Recipes
- Events
- Diagnostics
 - Trends
 - EtherCAT
 - System



Each page included in the template are setup as follows –

Home

A blank page to be used as the HMI's landing page. Recommended content suited for the Home page includes, but is not limited to, a Machine Overview, a Digital Twin, a Machine Image, or Metrics.

Application Specific Pages

To contain any control or information necessary to run specific parts of the machine.

Manual

SPT Framework v3/4 –

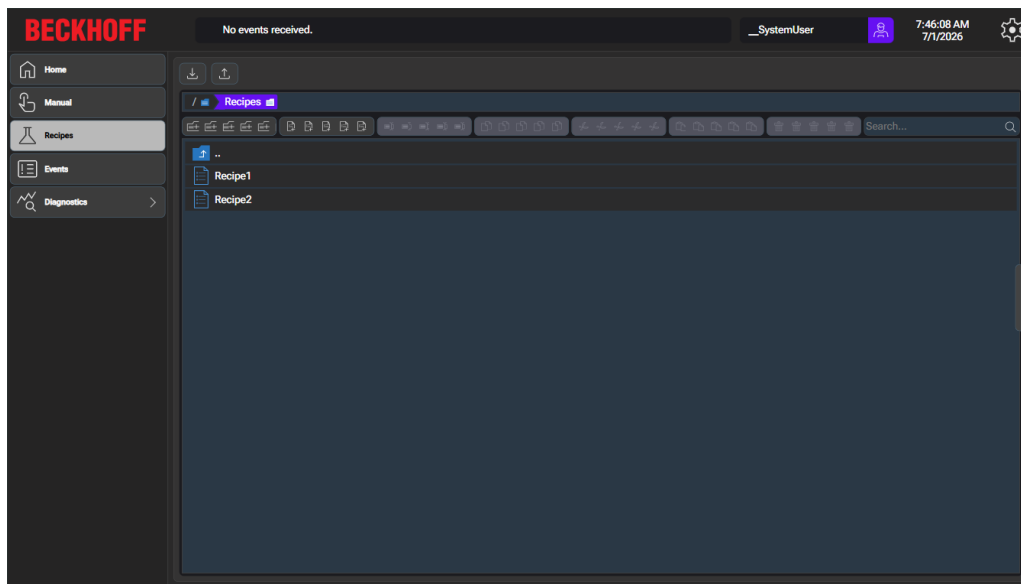
Empty by default but should be built out in collaboration between the PLC and HMI developers. Interaction and Data Entry that directly reflects the PLC project allowing manual operation of the machine during development, testing, and troubleshooting. A common example would be jog functionality for servo axes.

SPT Core –

Contains only a container with a function to automatically generate controls on Attached. User Controls must be created with names matching Control Types defined in the PLC (eg a Control Type AxisPTP needs a User Control called AxisPTPPanel).

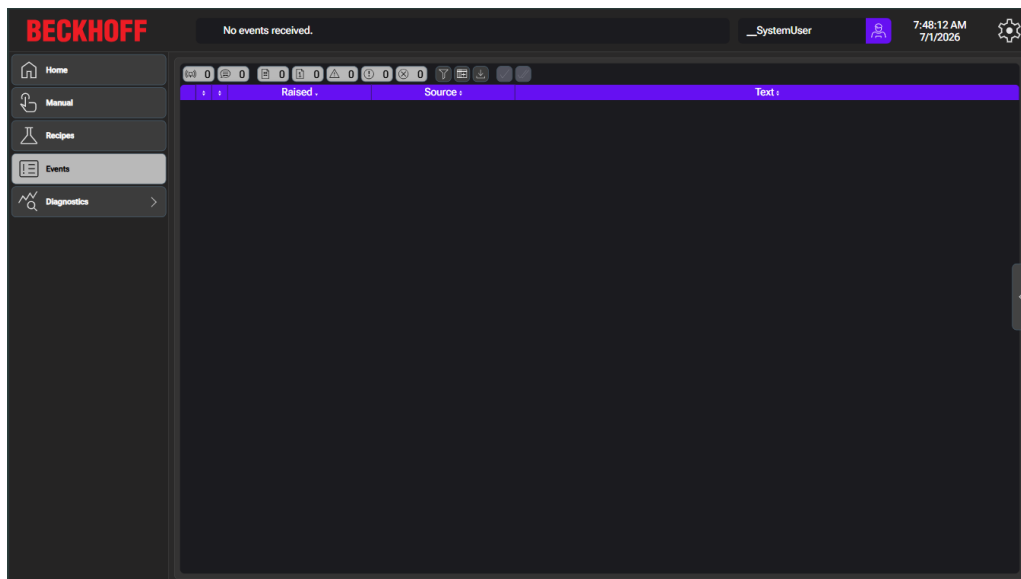
Recipes

Includes a recipe editor for adding or editing recipes, as well as download and upload buttons for backup and restore of recipe configuration.



Events

The events page has an Event Grid for viewing current and historical alarms. Default columns include Time Raised, Source, and Event Text. The grid's filter is set to events with domain equal to TcHmiEventLogger, severity $\geq$ Warning, and isActive true. Columns and Filters can be configured in the engineering environment and the client.

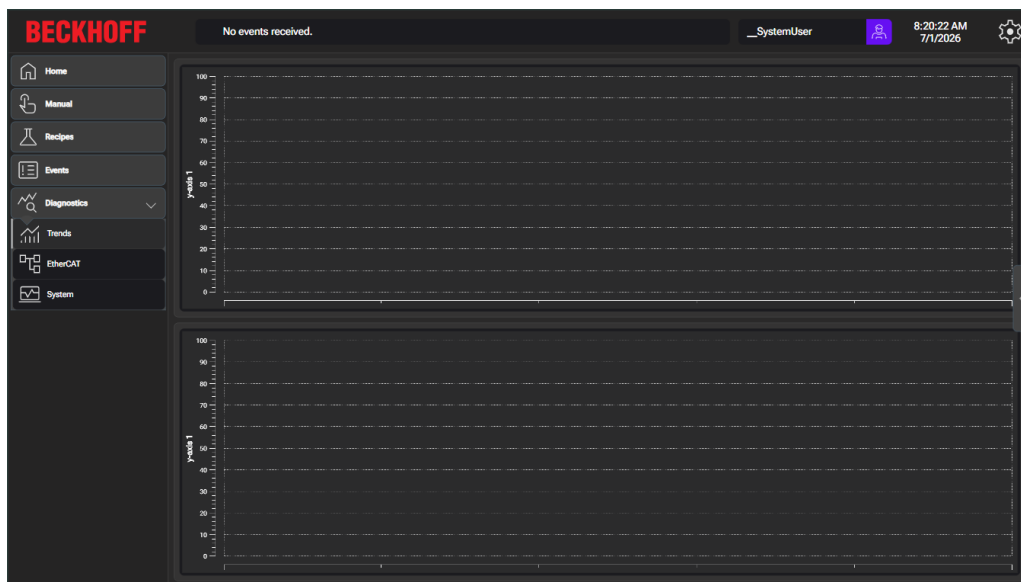


Diagnostics

The Diagnostics Navigation button is not configured to route to any page. Rather it expands to show the sub items below. However, if it makes sense in the current application to have a general Diagnostics page, it can be added here.

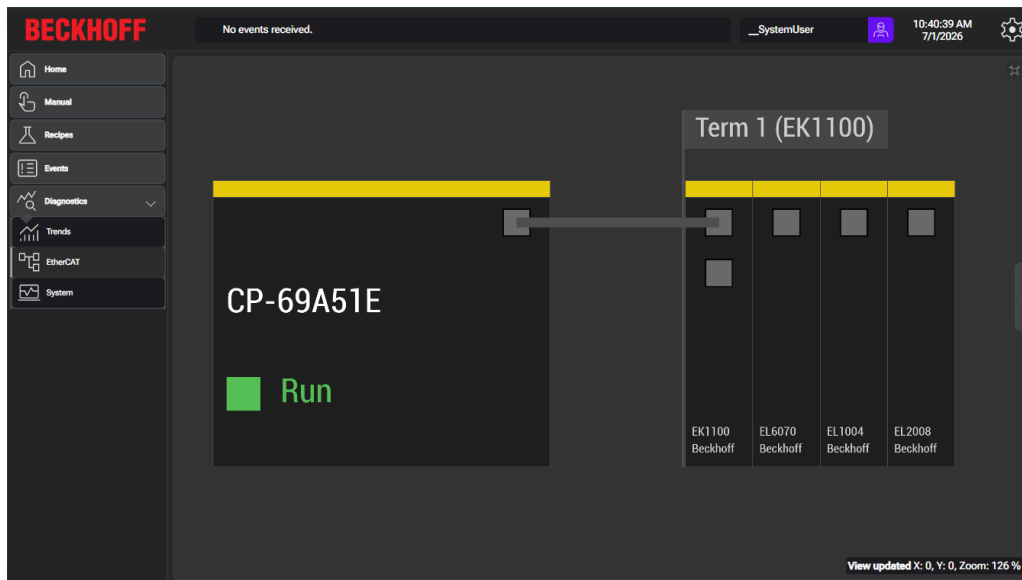
Trends

Includes minimally configured Trend Line Charts. This page is meant to be expanded as desired to contain multiple charts within a grid or tabbed navigation.



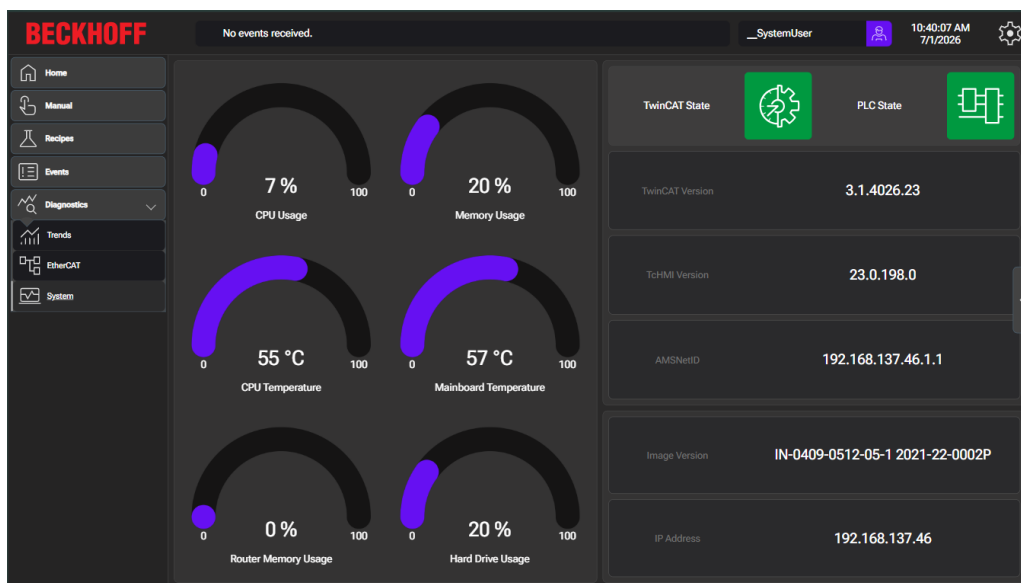
EtherCAT

Contains an overview of the configured EtherCAT network topology. It is fully interactable, and more info is displayed by clicking into devices. Inputs and Outputs can even be read/written, assuming the logged in user has the proper permissions.



System

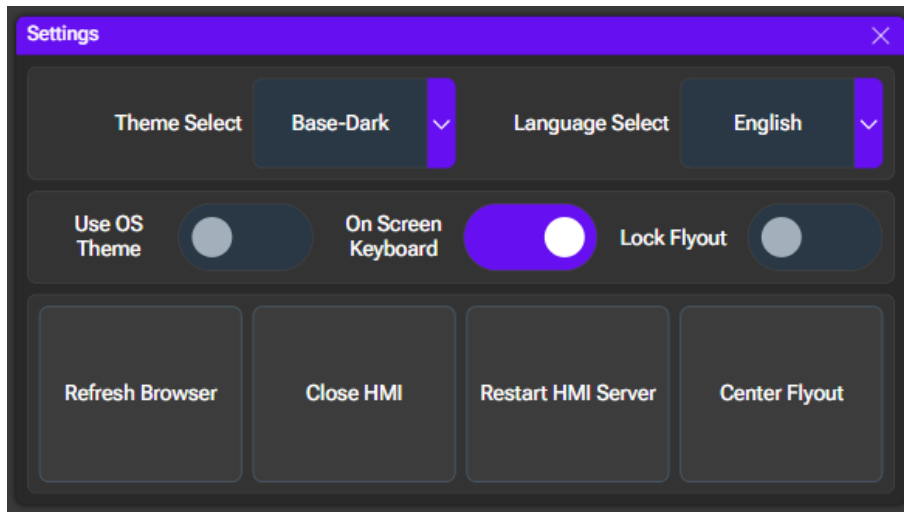
Common information which is useful for machine troubleshooting and receiving support.
Can be tailored to specific needs.



Settings Popup

Settings are in a popup accessed via a button in the header. Settings included are Theme (light and dark out of the box), Localization (if configured/not included in template), and buttons (with confirmation) to refresh the browser, close the HMI, and restart the HMI

Server. Quality of life functionality to re-center and lock positioning of PackML Flyout, and toggling usage of OS Theme and the on screen keyboard are also included.



Folder Layout

In addition to folders included in a blank TE2000 project, the following are also included –

- \Functions

- \Pages

- \Popups

 - \PanelPopups

- \Styling

- \UserControls

 - \Panels

Responsive Design

Where Responsive Design is required, elect to utilize the Grid and Flex Container controls. Use grids for more complex 2D layouts, and flex for linear 1D portions of a page.

Relative sizing and spacing should also be considered.

Grouping of Controls

Controls are grouped by grids, flex boxes and containers.

Common elements are grouped and nested together with the following hierarchy –

An outer Grid with 10px padding and 10px row and column gaps

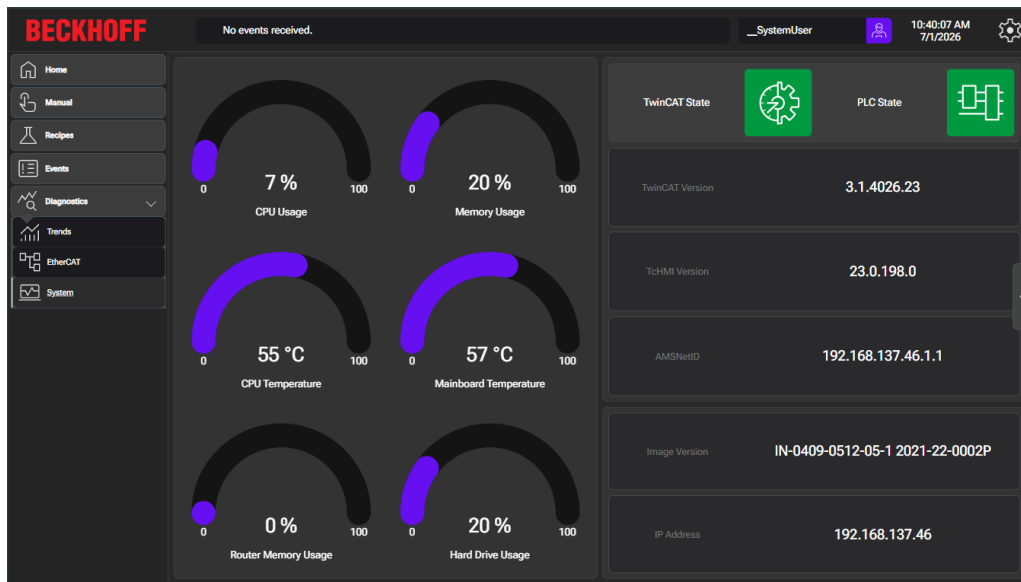
Inner Grids and/or Flex Boxes with Left/Right/Width/Height set to 0px/0px/100%/100%, respectively. Grid and Flex layouts have 10px rows/column gaps. Any further nested Grids or Flex Boxes following this rule as well.

Individual controls should be Left/Right 0px/0px. For grids, Width and Height will both be 100%. For flex, if the flex direction is row, height should be 100%. If the flex direction is column, width should be 100%. The width/height that is not 100% should be set to some reasonable guiding value, but the real sizing is affected by the `FlexItemLayout>Flex` property. By default, most are 1 1 auto, which allows controls to grow and shrink as needed. If a control should be bigger than others in the flex box, increase the first value, if it should be smaller, increase the second value.

See Technical Guide for more details on how to do this.

Items are grouped together using a background color and border depicting the grouping.

In general, elements should not butt up against each other and should have consistent space between them. See the template's System page as an example.



Styling

In this section, default styling is discussed. In general, custom styling is done in .theme files first, theme specific CSS files second, and global CSS files last.

Control Default Styling

Rounded corners: 0.45rem (rem = relative to root element's font size, for scalability)

Background colors: Solid Colors, no gradients

Borders: Use sparingly and intentionally for delineating groups of elements. Should be subtle

Drop Shadows: Do not use

Colors

Colors should be rare and intentional

Grouping containers separate related controls, using background color and borders to do so

Light theme:

Background: #80FFFFFF / rgba(255, 255, 255, 128)

Border: #DCDCDC / rgb(220, 220, 220) 1px

Dark theme:

Background: #0CFFFFFF / rgba(255, 255, 255, 12)

Border: #464646 / rgb(70, 70, 70) 1px

Slight darkening of navigation and header is used to help differentiate the different sections with a faint border color

Light Theme:

Background: #80F6F7F8 / rgba(246, 247, 248, 128)

Border: #DCDCDC / rgba(220, 220, 220, 255) 1px

Dark Theme:

Background: #252424 / rgba(37, 36, 36, 255)

Border: #464646 / rgba(70, 70, 70, 255) 1px

Highlight/Primary: This is typically the company's primary color. Used to highlight imported items that need attention.

Default highlight color: #6610F2 / rgb(102, 16, 242)

Changed in theme CSS files via :root selector



Semantic Coloring for alarms, buttons, status indicators, etc. –

Good/Start: #51D777 / rgb(81, 215, 119) (Green)



Warning: #FBA712 / rgb(251, 167, 18) (Orange)



Critical/Stop: #EF4949 / rgb(239, 73, 73) (Red)



Information/Reset: #175DB8 / rgb(23, 93, 184) (Blue)



Idle/Inactive/Neutral: #B9B9B9 / rgb(185, 185, 185) (Grey)



Font

Roboto Flex font family is used with a 1.8cqh font size. However, font size is set to not go smaller than 10px, for readability.

Roboto Flex upgrades Roboto so it becomes a more powerful typeface system. Roboto Flex is a variable font that you can customize to express and finesse your text in ways never before possible. Using a relatively new version of the OpenType file format, variable fonts allow one file to contain multiple stylistic variations, so the final file size of the complete package is remarkably small, given the range of expressive styles now available.

If additional font sizes are needed, four should be used at most. For example –

H1, H2, H3, Body

Iconography

Icons are utilized for quick visualization of functionality, and their design must be consistent. To this end, [Google Material Font Icons](#) were chosen to be used throughout the HMI. If text is to be omitted and only an icon is used, the icon should be as universally and immediately recognizable as possible.

Stroke weight is 200 and color is greyscale.

Icons are located in the \Images\Icons folder

Icons utilized include –

	Home
	Manual
	Recipes
	Events
	Diagnostics
	Trends
	EtherCAT
	System

	Settings
	Info
	Start
	Stop
	Reset
	Download
	Upload

UI Components

Certain UI Components follow specific practices throughout a project. This section describes a few such components.

Standard Button Setup and Databinding

Push Button - No latching, use State Symbol property of the control for momentary state change.

Toggle Button – Use the State Symbol property of the control.

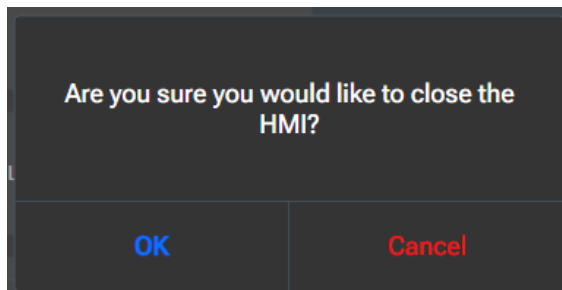
Toggle Switch - Add a label to the left side.

Confirmation Dialog

A standard Confirmation Dialog is used anywhere confirmation is required when pressing a button to initiate an action.

The dialog has a static width and height, and the message should not exceed 100 characters.

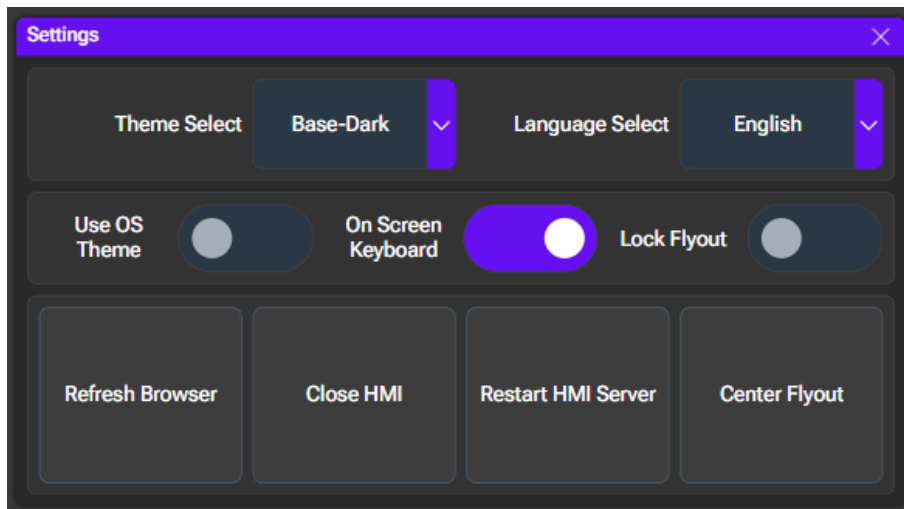
Buttons span the full width - OK on the left in blue and Cancel on the right in red. The background color of the buttons match the background color of the popup.



Popups

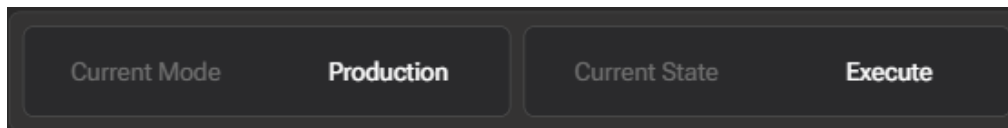
Popups have a header colored using the primary/highlight color. The header must contain a succinct title describing what the popup is for. The border and background colors match.

The main grid in a popup's .content or .usercontrol file does not adhere to the 10px padding rule, as the popup element already provides a 10px padding.



Data Display Tiles

Read only data must not be displayed in inputs, even if said input is disabled. The Data Display Tile control has been created for displaying data with a label. Making the data stand out from the label, while not being confused as an input, is the intent of this control. Font color and size of label and value are independently configurable, but the value should have more contrast in color.



Setup and Conventions

Best practices and guidelines for project development are outlined here.

Naming Conventions

Control IDs:

[location]-[what it is]-[what it does]

Functions:

Pascal Case - what it does

Variables:

Camel Case - what it does - verbose

Version Control

The .gitignore is modified to include *.nuget

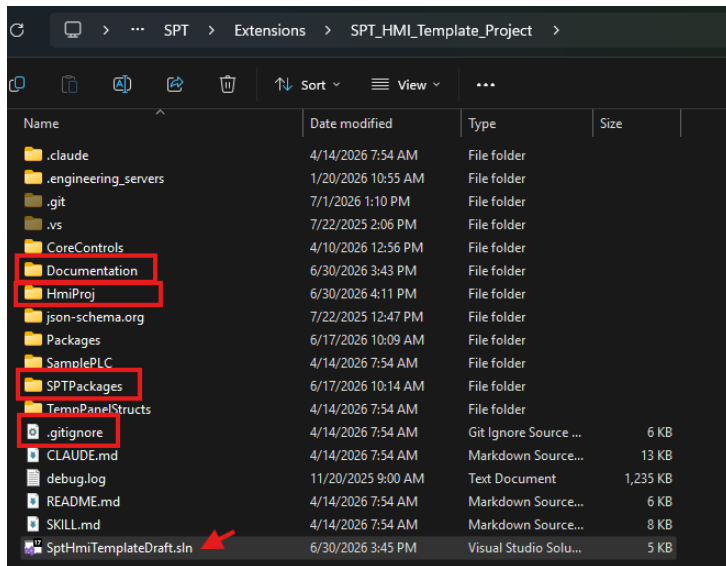
Any framework control's .nupkg files will be in the \SPTPackages folder of the project's parent directory.

Custom Packages should be copied from the \SPTPackages folder to C:\ProgramData\Beckhoff\NuGetPackages on the engineering PC.

New Project Generation

New projects will always start by copying the template fresh from the repo.

Create a solution file, copy the .gitignore file, HmiProj, SPTPackages, and optionally the Documentation folders to be on the same folder level as the .sln file.



Make sure to have the latest nuget packages from the SPTPackages folder (copied to C:\ProgramData\Beckhoff\NuGetPackages). Then open the .sln, Add existing, and choose the .hmiproj file in the HmiProj Folder

